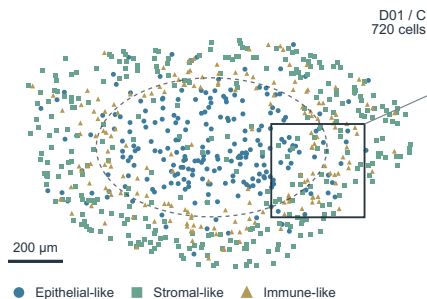
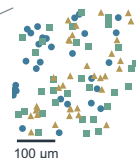


a Spatial compartments



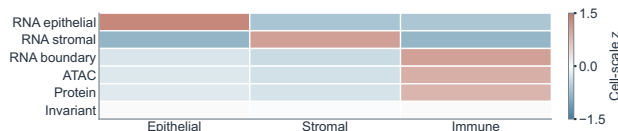
b Same-field ROI



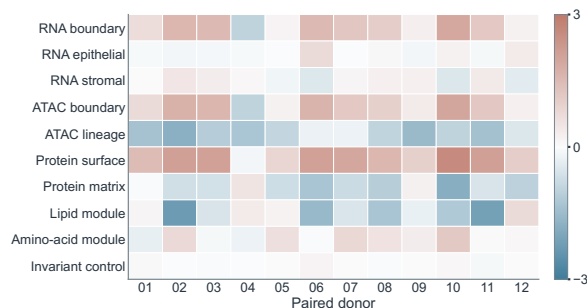
c Boundary RNA



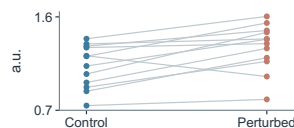
d State-resolved molecular profiles



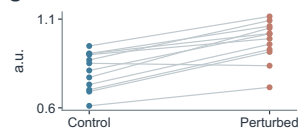
e Paired multiomic changes



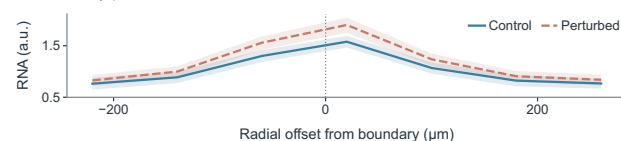
f RNA boundary



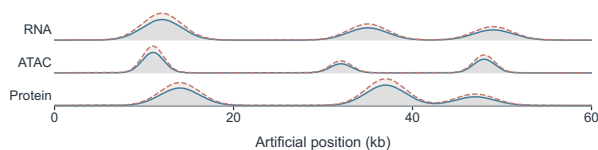
g Protein surface



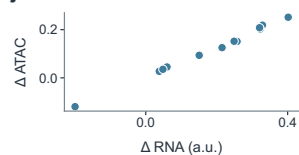
h Boundary profile



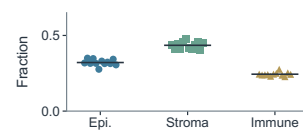
i Illustrative locus tracks



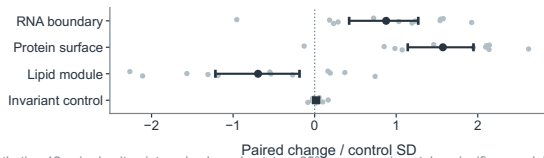
j Cross-assay pairing



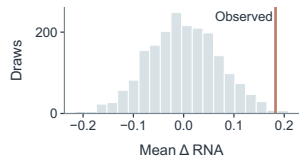
k State composition



l Specificity and direction



m Paired sign-flip null



n Boundary-width sensitivity

